

# CIS 8693 – The 50K Group Report

## Project Topic: Adult Census Income

### 1. Problem & Goal

The goal of this project is to predict whether an individual earns more than \$50,000 per year based on demographic, educational, and employment-related characteristics. By identifying the factors most strongly associated with higher income levels, financial institutions can better understand income distribution patterns and support marketing initiatives.

#### 1.1 Objectives:

- Understand the structure and features of the dataset.
- Prepare, clean, and preprocess the data.
- Handle missing values and remove redundant variables.
- Explore relationships between demographic, educational, and employment characteristics and income level.
- Apply multiple predictive models, including Logistic Regression, Decision Tree, Random Forest.
- Compare model performance using classification metrics.
- Identify potential overfitting issues and improve model generalization when necessary.
- Select the best-performing model based on testing performance.
- Generate insights regarding the factors associated with income levels above and below \$50K.

### 2. Data Description

The Adult Census Income dataset was used in this project. The dataset contains demographic, educational, and employment-related information for individuals and is commonly used for classification problems involving income level.

The dataset contains:

- 32,561 observations

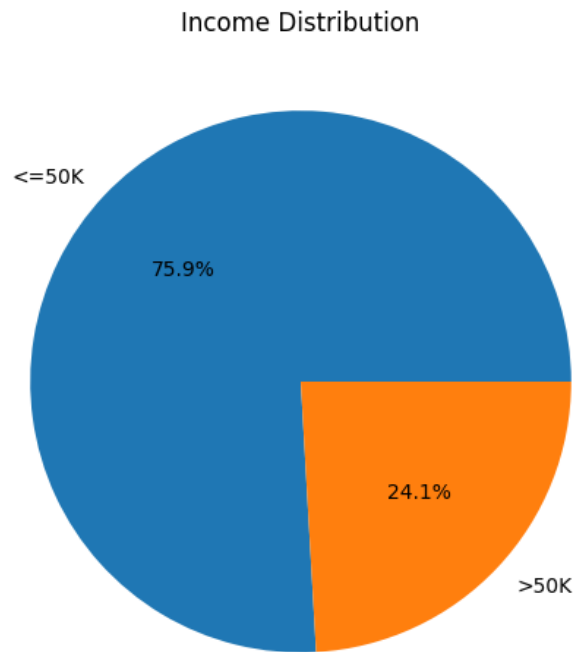
- 15 original variables
- One dependent variable (income)
- Multiple independent variables

## 2.1 Dependent Variable: Income

This variable classifies individuals into two categories:

- $\leq 50K$ : Annual income is less than or equal to \$50,000
- $> 50K$ : Annual income is greater than \$50,000.

The distribution of the target variable is:



Approximately 75.9% of individuals earn \$50,000 or less, while 24.1% earn more than \$50,000 annually. The target variable has a **moderate class imbalance**.

## 2.2 Independent Variables

The dataset contains both numerical and categorical features. The original variables are:

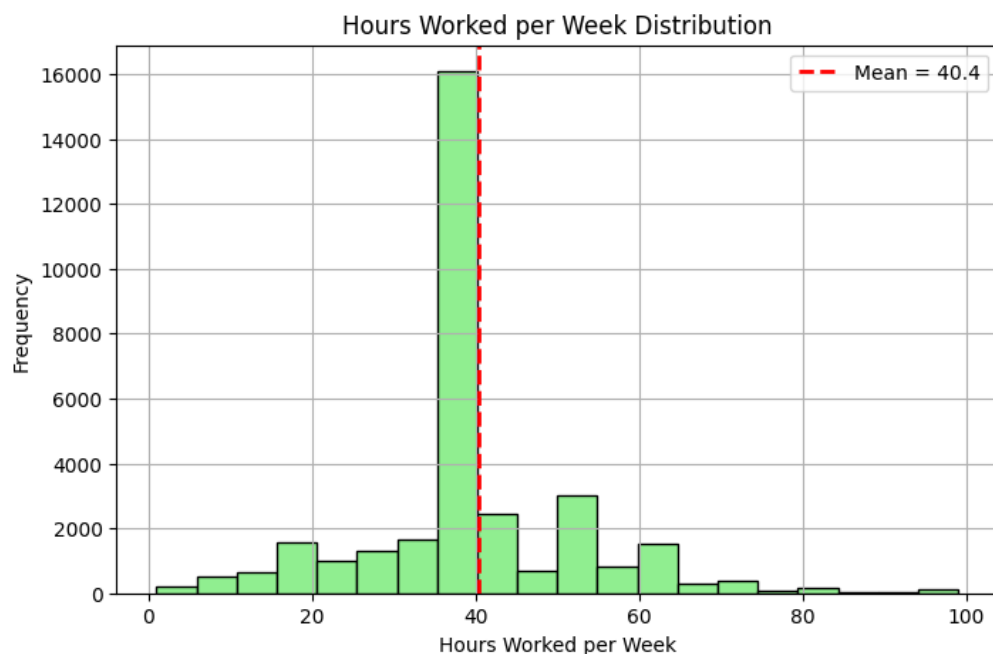
|                     |                       |
|---------------------|-----------------------|
| Numerical Variables | Categorical Variables |
|---------------------|-----------------------|

|                |                |
|----------------|----------------|
| age            | workclass      |
| fnlwgt         | education      |
| education.num  | marital.status |
| capital.gain   | occupation     |
| capital.loss   | relationship   |
| hours.per.week | race           |
|                | sex            |
|                | native.country |

Descriptive statistics were calculated for some of the most relevant numerical variables in the dataset. The results provide an overview of the population represented in the data:

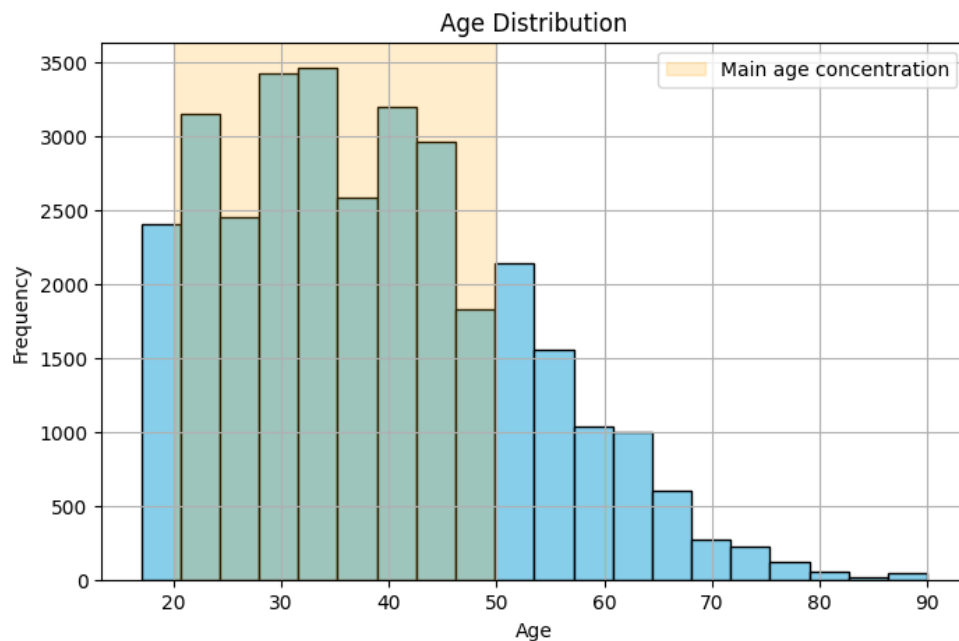
| index | age       | capital.gain | capital.loss | hours.per.week |
|-------|-----------|--------------|--------------|----------------|
| count | 32,561.00 | 32,561.00    | 32,561.00    | 32,561.00      |
| mean  | 38.58     | 1,077.65     | 87.30        | 40.44          |
| std   | 13.64     | 7,385.29     | 402.96       | 12.35          |
| min   | 17.00     | -            | -            | 1.00           |
| max   | 90.00     | 99,999.00    | 4,356.00     | 99.00          |

Individuals work an **average** of **40.4 hours per week**, with a **median** of **40 hours**. This suggests that most individuals in the dataset work approximately a standard **full-time schedule**:

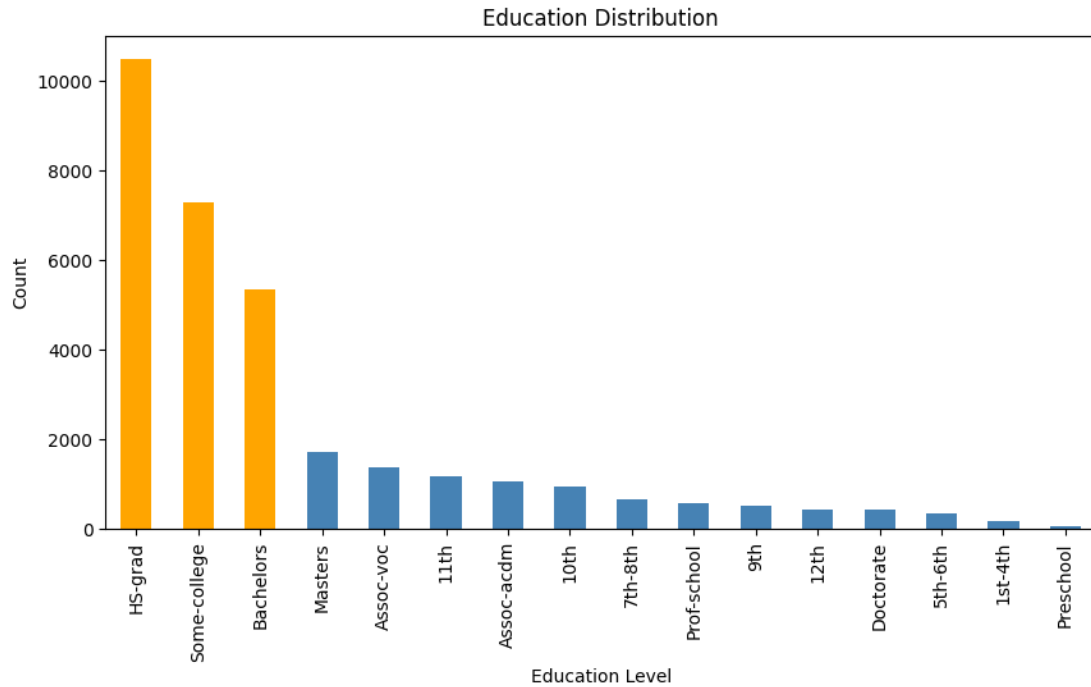


The variables `capital.gain` and `capital.loss` show a wide range of values. While many individuals report no gains or losses, some observations contain very large values, indicating **substantial variation** in financial characteristics across the population.

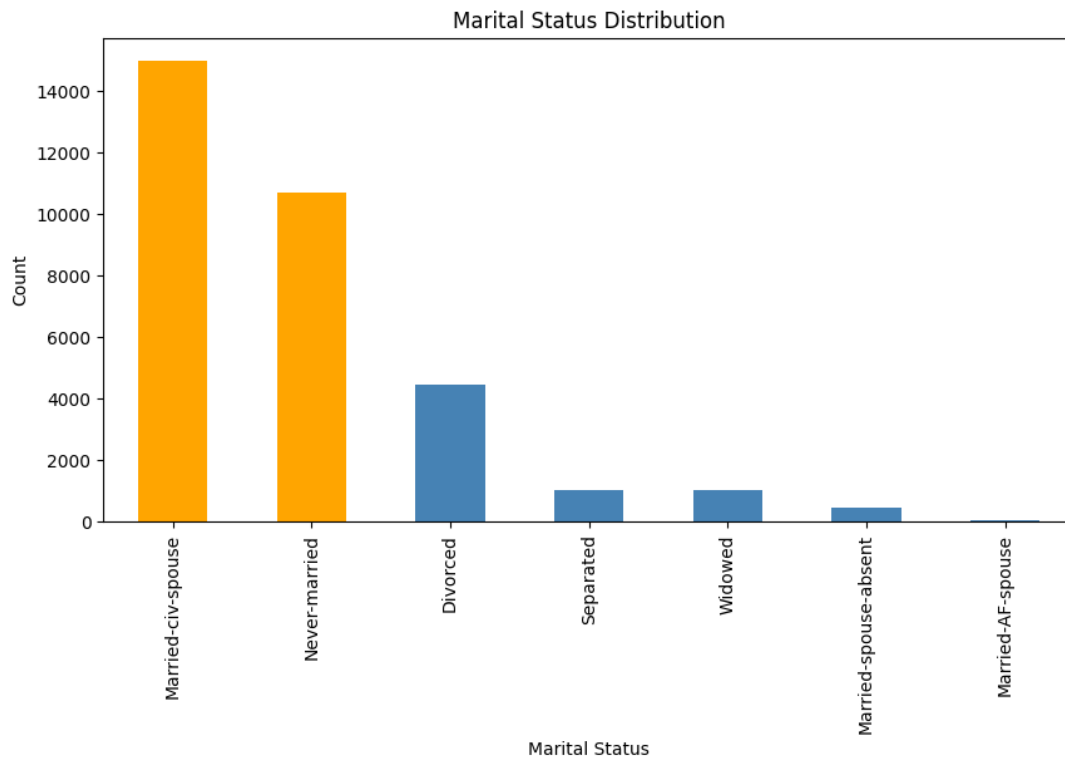
The **average age** of individuals in the dataset is **38.6 years**, with ages ranging from 17 to 90 years. Most observations are concentrated between **28 and 48 years old**, indicating that the dataset primarily represents the working-age population:



The **largest education category** in the dataset is HS-grad, followed by Some-college and Bachelors, indicating that most individuals have completed at least secondary education:



The **marital status distribution** shows that the largest groups in the dataset are Married-civ-spouse and Never-married. This indicates that the dataset contains a substantial representation of both married and unmarried individuals, providing diversity in household and family structures:



### 3. Data Cleaning

The dataset contains multiple elements that should be handled before applying predictive models.

#### 3.1 Column names

Several column names contained dot symbols (.). This feature can generate potential problems during data preprocessing and model implementation, so to solve this issue the symbols were replaced with underscores (\_).

#### 3.2 Missing Values

After analyzing the whole dataset some records show the symbol ? as values. These records were identified in three categorical variables:

- workclass: 1,836 missing values
- occupation: 1,843 missing values
- native\_country: 583 missing values

The symbol ? was converted to **NaN** for better handling these missing values. The total amount of values was **4,262**, representing approximately **0.87%** of the dataset. To preserve all observations, missing values were imputed using the **mode** (most frequent category) of each variable.

#### 3.3 Duplicate Records

Only 24 **duplicate records** were identified during the data cleaning process and were removed from the dataset.

#### 3.4 Unnecessary or redundant variables

The variables **education** and **fnlwgt** were removed. The variable education\_num contains the same information as education in numeric form, making education redundant. The fnlwgt represents a sampling weight created by the U.S. Census Bureau. It indicates how many people in the overall population are represented by a particular observation in the

dataset. Thus, `fnlwgt` does not describe the person directly and does not provide business insight.

### 3.5 Data Preparation

The target variable income was converted into a **binary format** before running the models. It was represented numerically with:

- 0 = income less than or equal to \$50K
- 1 = income greater than \$50K

Several variables in the Adult Census dataset were categorical, including `workclass`, `marital_status`, `occupation`, `relationship`, `race`, `sex`, and `native_country`. Categorical variables were **converted into numeric category codes** using the pandas categorical data type (`cat.codes`). This transformation allowed machine learning algorithms to process non-numeric attributes while preserving the information contained in each category.

## 4. Modeling

To predict whether an individual earns more than \$50,000 per year, three classification models were evaluated: Logistic Regression, Decision Tree and Random Forest.

Numerical features were **scaled** using `StandardScaler` before training the Logistic Regression model. This process transforms variables to a common scale with a mean of zero and a standard deviation of one. This prevents variables with larger numeric ranges from exerting greater influence on the model and improves training stability. Tree-based models, such as Decision Trees and Random Forests, were trained using the original feature values because they are generally insensitive to feature scaling.

As it was said before, the target variable exhibited **class imbalance**, with a substantially larger proportion of individuals earning \$50,000 or less than those earning more than \$50,000. To address this issue, all classification models were trained using the parameter **`class_weight='balanced'`**. This approach automatically adjusts the importance of each class according to its frequency in the dataset, helping the models better identify observations belonging to the minority income group.

Before model training, the dataset was split into training and testing subsets using the **75/25 rule**. The training dataset was used to fit the models, while the testing dataset was used to evaluate their performance on unseen observations.

#### 4.1 Logistic Regression

Logistic Regression served as a baseline classification model. The algorithm estimates the probability that an individual belongs to the income class above \$50,000 based on the predictor variables.

The results obtained for the training data:

| Confusion Matrix (Accuracy 0.7719) |        |       |
|------------------------------------|--------|-------|
| Actual/Predicted                   | 0      | 1     |
| 0                                  | 14,274 | 4,196 |
| 1                                  | 1,371  | 4,561 |

The results obtained for the testing data:

| Confusion Matrix (Accuracy 0.7687) |       |       |
|------------------------------------|-------|-------|
| Actual/Predicted                   | 0     | 1     |
| 0                                  | 4,797 | 1,431 |
| 1                                  | 451   | 1,456 |

The model produced very similar training and testing accuracies, indicating good generalization and limited evidence of overfitting.

#### 4.2 Decision Tree

Decision Trees classify observations through a sequence of hierarchical decision rules. The model is particularly useful because its decision-making process can be visualized and interpreted.

The results obtained for the training data:

| Confusion Matrix (Accuracy 0.9784) |        |       |
|------------------------------------|--------|-------|
| Actual/Predicted                   | 0      | 1     |
| 0                                  | 18,385 | 85    |
| 1                                  | 443    | 5,489 |

The results obtained for the testing data:

| Confusion Matrix (Accuracy 0.8166) |       |       |
|------------------------------------|-------|-------|
| Actual/Predicted                   | 0     | 1     |
| 0                                  | 5,478 | 750   |
| 1                                  | 742   | 1,165 |

The large difference between training accuracy (97.84%) and testing accuracy (81.66%) suggests that the model learned patterns specific to the training data and **did not generalize well** to unseen observations.

The large difference between training and testing accuracy suggested **overfitting**. To improve model generalization, the **maximum tree depth** was limited to ten levels.

#### 4.3 Tuned Decision Tree Results (max\_depth = 10)

The results obtained for the training data:

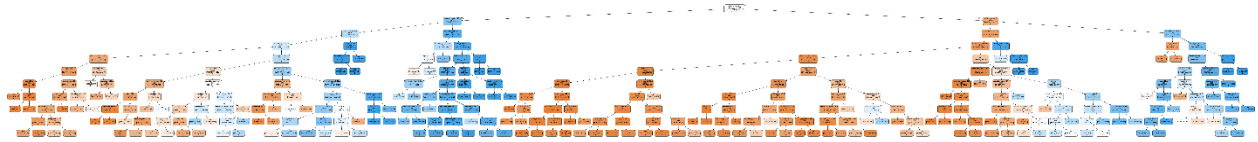
| Confusion Matrix (Accuracy 0.8116) |        |       |
|------------------------------------|--------|-------|
| Actual/Predicted                   | 0      | 1     |
| 0                                  | 14,658 | 3,812 |
| 1                                  | 785    | 5,147 |

The results obtained for the testing data:

| Confusion Matrix (Accuracy 0.7966) |       |       |
|------------------------------------|-------|-------|
| Actual/Predicted                   | 0     | 1     |
| 0                                  | 4,865 | 1,363 |
| 1                                  | 292   | 1,615 |

After limiting the maximum tree depth to ten levels, the model achieved a testing accuracy of 79.66%, compared to 81.66% for the original unrestricted tree. Although overall accuracy decreased slightly, the difference between training accuracy (81.16%) and testing accuracy (79.66%) became substantially smaller. This indicates that the tuned model generalized better on unseen observations and successfully reduced overfitting.

The following image shows the tree structure:



The resulting tree revealed that income classification depends on a **combination** of demographic, educational, employment, and financial factors. Variables such as capital gain, education level, age, occupation, marital status, and hours worked per week contributed to the classification process through multiple decision paths.

4.4 Random Forest

A Random Forest classifier was applied as a collection learning technique that combines the predictions of multiple decision trees. The model was configured with 500 decision trees in order to improve prediction stability and reduce variance.

The results obtained for the training data:

| Confusion Matrix (Accuracy 0.8411) |        |       |
|------------------------------------|--------|-------|
| Actual/Predicted                   | 0      | 1     |
| 0                                  | 15,176 | 3,294 |
| 1                                  | 583    | 5,349 |

The results obtained for the testing data:

| Confusion Matrix (Accuracy 0.8182) |       |       |
|------------------------------------|-------|-------|
| Actual/Predicted                   | 0     | 1     |
| 0                                  | 5,023 | 1,205 |
| 1                                  | 274   | 1,633 |

Random Forest achieved the **highest overall testing performance** among the evaluated models. It obtained the strongest Recall and ROC-AUC values while maintaining competitive Accuracy. These results indicate that the model was more effective at identifying individuals belonging to the higher-income category while preserving overall classification performance.

5. Model Selection

While confusion matrices give a detailed view of model predictions and classification errors, additional evaluation metrics were used to facilitate a direct comparison among models. Furthermore, the dataset presented a clear **imbalance** between the >50K income level earners and <50K income. Because all these reasons, Accuracy, Precision, Recall, F1 and ROC-AUC metrics were selected because they provide complementary perspectives on classification performance.

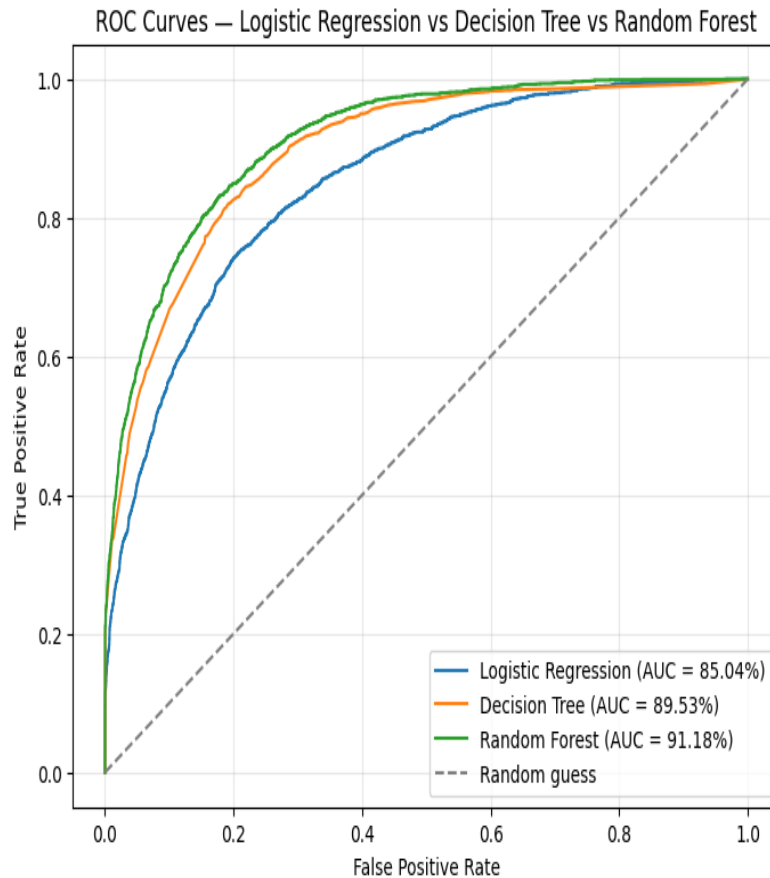
Each metric shows a different aspect of model performance:

- **Accuracy:** Measures the proportion of correctly classified observations.
- **Precision:** Measures the percentage of predicted high-income individuals who actually belong to the high-income group.
- **Recall:** Measures the percentage of actual high-income individuals correctly identified by the model.
- **F1 Score:** Measures a classification model's accuracy by finding the harmonic mean of precision and recall.
- **ROC-AUC:** Measures the model's ability to distinguish between income classes across different classification thresholds.

The following table summarizes the model results:

| Model                | Accuracy      | Precision     | Recall        | F1            | ROC-AUC       |
|----------------------|---------------|---------------|---------------|---------------|---------------|
| Logistic Regression  | 76.87%        | 50.43%        | 76.35%        | 60.74%        | 76.69%        |
| Decision Tree        | 81.66%        | 60.84%        | 61.09%        | 60.96%        | 74.52%        |
| Tuned Decision Tree  | 79.66%        | 54.23%        | 84.69%        | 66.12%        | 81.40%        |
| <b>Random Forest</b> | <b>81.82%</b> | <b>57.54%</b> | <b>85.63%</b> | <b>68.83%</b> | <b>83.14%</b> |

The following graph visualizes the ROC-AUC results for each model:



Although the Tuned Decision Tree achieved higher Precision, Random Forest achieved superior Accuracy, Recall, and ROC-AUC values. Since the objective was to identify individuals earning more than \$50,000, **Random Forest** provided the strongest overall performance and was selected as the final model.

## 6. Managerial Insights

### 6.1 Education

Education level emerged as one of the strongest factors of income level. Individuals with higher levels of education, like college and advanced degrees, were considerably more likely to belong to the income group earning more than \$50,000 annually. This suggests that educational achievement remains an important factor associated with income.

### 6.2 Age

Income levels are likely to increase during mid-career years. Individuals between 35 and 50 years of age were more likely to belong to the higher-income category, indicating that work experience and career progression play an important role in income growth. This pattern is consistent with the accumulation of work experience, professional development, and career advancement opportunities over time.

### 6.3 Working hours

Individuals working more hours per week were generally more likely to earn above \$50,000. This finding suggests that labor-force participation and work intensity are associated with higher earnings.

### 6.4 Capital Gains

Capital gains were clearly associated with the higher-income group. Individuals reporting investment-related income were considerably more likely to exceed the \$50,000 income, highlighting the importance of financial assets as an indicator of earning level.

### 6.5 Marital Status

Marital status was identified as an important predictor within the classification models. Individuals in the Married-civ-spouse category were more frequently associated with income levels above \$50,000, suggesting that household and family characteristics may be related to earning potential.

### 6.6 Recommendations

Based on the results of this project, financial institutions can use demographic, educational, and employment-related information to improve customer **targeting strategies**. The **Random Forest** model demonstrated the strongest ability to identify individuals likely to earn more than \$50,000 and can be used as a detective tool for marketing campaigns involving premium financial services.

Because the model's precision remains the weak point, it should be used to **prioritize and rank potential customers** rather than as a final decision-making tool. Approximately four out of ten individuals predicted to belong to the >\$50K income group may not actually fall into that category.

Different **probability thresholds** may be evaluated depending on marketing objectives. Lower thresholds may increase the number of potential customers identified, while higher thresholds may improve targeting accuracy.

Since the Adult Census dataset was collected in 1994, future implementations should be validated using more **recent data**. The model should also be periodically monitored and updated to account for changes in demographics, employment patterns, and income distributions over time.

## 7. Conclusion

This project developed and evaluated multiple classification models to predict whether an individual's annual income exceeds \$50,000. After understanding and cleaning the dataset, categorical encoding, columns standardization, and class balancing, Logistic Regression, Decision Tree, and Random Forest models were evaluated.

The results demonstrated that **Random Forest** provided the strongest overall performance, achieving the highest metrics. Furthermore, the analysis identified education level, age, marital status, hours worked per week, and capital gains as important factors associated with higher income levels.

These findings suggest that machine learning techniques can provide valuable support for **customer segmentation** and targeted marketing initiatives within financial institutions.

These results demonstrate the practical value of machine learning techniques for income classification problems and highlight their potential application in customer targeting and market segmentation strategies.

### Dataset source:

- <https://www.kaggle.com/datasets/uciml/adult-census-income?resource=download>